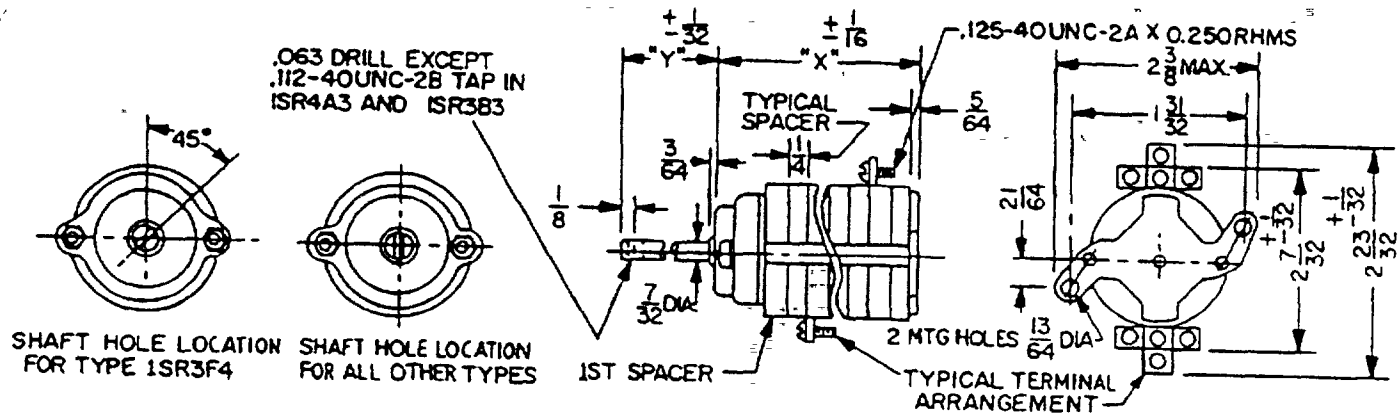
INCH-POUND
MIL-S-15291/3A(SH)
22 July 1991
SUPERCEDING
MIL-S-15291/3(SH)
17 April 1981

MILITARY SPECIFICATION SHEET

SWITCHES, ROTARY, SNAP ACTION CLASS 1SR BASE MOUNTED, SIDE CONNECTED

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-15291.



DETAIL "A"

NOTES
1 Dimensions are in inches.
2 Unless otherwise specified, tolerance is plus or minus 0.015

FIGURE 1. *Class 1SR switched, base mounted.*

AMSC N/A

FSC 5930

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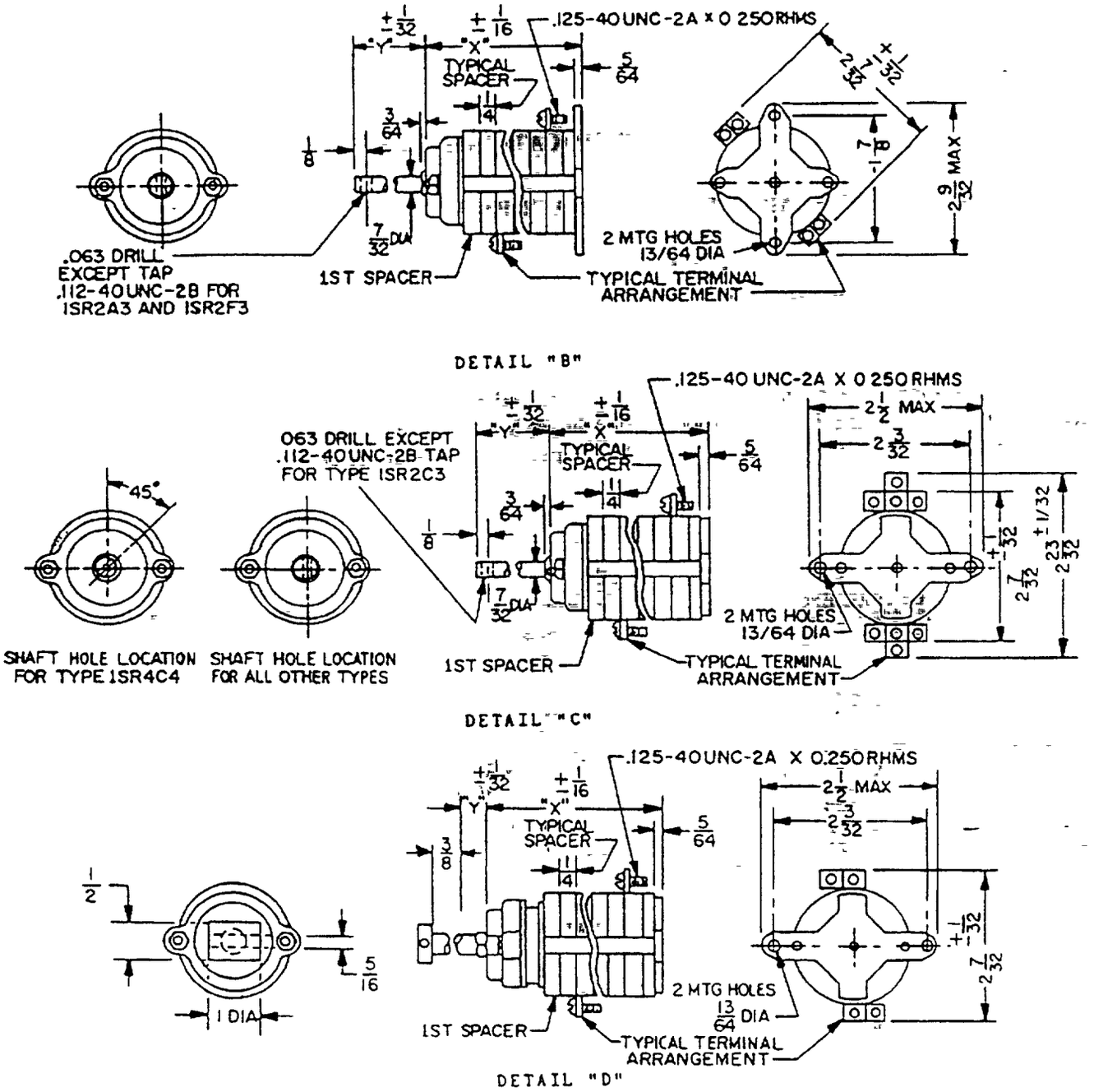


FIGURE 1. *Class 1SR switched, base mounted — Continued*

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TABLE I. *Type and switching characteristics.*

M15291/3 Dash No Type designation	Detail ref	Dim "X"	Handle or shaft position	Circuit and spacer configuration rotor position, spacer locations, terminal marking and locations									Notes
	Torque in-lbs	Dim "Y"		No 1	No 2	No 3	No 4	No 5	No 6	No 7	No. 8	No. 9	
-001	Fig 1 detail B	1 3/8	ON OFF ON		1 2	3 4							
ISR2A3	4	1 5/32											
-002	Fig 1 detail B	1 3/8	ON OFF ON		1 2	3 4							
ISR2A4	4	9/32											
-003	Fig 1 detail D	1 9/16	ON OFF		1 2	3 4							
ISR2A4A	4	9/32											
-004	Fig 1 detail C	1 5/8	ON OFF ON		1 2	3 4	5 6						
ISR3A4	4	9/32											
-005	Fig 1 detail A	2 1/8	ON OFF ON	1/2 Blank		1 2	3 4	5 6	7 8	1/2 Blank			
ISR4A3	4	1 5/32											
-006	Fig 1 detail A	2 1/8	ON OFF ON	1/2 Blank		1 2	3 4	5 6	7 8	1/2 Blank			
ISR4A4	4	9/32											
-007	Fig 1 detail D	2 5/16	OFF ON	1/2 Blank		1 2	3 4	5 6	7 8	1/2 Blank			
ISR4A4A	4	9/32											
-008	Fig 1 detail A	1 3/8	ON 2 OFF ON 1		1 2 3	4 5 6							
ISR2B4	4	9/32											

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TABLE I *Type and switching characteristics – Continued*

MIL-S-15291/3 dash no. Type designation	Detail ref. Torque in-lbs	Dim "X" Dim "Y"	Handle or shaft position	Circuit and spacer configuration rotor position, spacer locations, terminal marking and locations No 1	No 2	No 3	No 4	No 5	No 6	No 7	No. 8	No 9	Notes
-009 ISR2B4A	Fig 1 detail A / 4	$2\frac{1}{8}$ / $\frac{9}{32}$	2, OFF, OFF, 1	1/2 Blank		1, 2, 3	4, 5, 6		1/2 Blank				
-010 ISR3B3	Fig 1 detail A / 4	$2\frac{5}{8}$ / $1\frac{5}{32}$	2, OFF, OFF, 1	1/2 Blank		1, 2, 3	4, 5, 6		7, 8, 9			1/2 Blank	
-011 ISR3B4	Fig 1 detail A / 4	$2\frac{5}{8}$ / $\frac{9}{32}$	2, OFF, OFF, 1	1/2 Blank		1, 2, 3	4, 5, 6		7, 8, 9			1/2 Blank	
-012	Fig 1 detail A / 4	$2\frac{5}{8}$ / $\frac{9}{32}$	2, OFF, OFF, 1	1/2 Blank		1, 2, 3	4, 5, 6		7, 8, 9		10, 11, 12	1/2 Blank	
-013 ISR1C4	Fig 1 detail B / 4	$1\frac{3}{8}$ / $\frac{9}{32}$	2, 2		1, 2, 3, 4								_1
-014 ISR2C3	Fig 1 detail C / 4	$1\frac{7}{8}$ / $1\frac{5}{32}$	2, 2		1, 2, 3, 4			5, 6, 7, 8					_1
-016 ISR2C4A	Fig 1 detail C / 4	$1\frac{5}{8}$ / $\frac{9}{32}$	2, 2		1, 2, 3, 4		5, 6, 7, 8						_1
-017 ISR3C4	Fig 1 detail C / 4	$2\frac{1}{8}$ / $\frac{9}{32}$	2, 2		1, 2, 3, 4		5, 6, 7, 8		9, 10, 11, 12				_1

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TABLE I. Type and switching characteristics – Continued

MIL5291/3 Dash No Type designation	Detail ref / Torque in-lbs.	Dim "X" / Dim "Y"	Handle or shaft position	Circuit and spacer configuration rotor position, spacer locations, terminal marking and locations No. 1	No. 2	No. 3	No. 4	No. 5	No. 6	No. 7	No. 8	No. 9	Notes
-018 ISR4C4	Fig 1 detail C / 8	$2\frac{3}{4}$ / $\frac{9}{32}$	2 1 1 2		1 2 4 3		5 6 8 7		9 10 12 11		13 14 16 15		[1]
-019 ISR2E4	Fig 1 detail A / 4	$2\frac{1}{8}$ / $\frac{9}{32}$	2 OFF 1	1/2 Blank	1	2 4 3	5	6 8 7	1/2 Blank				
-020 ISR2F3	Fig 1 detail B / 4	$1\frac{3}{8}$ / $1\frac{5}{32}$	2 OFF 1 2 1		1 2	4 3							
-021 ISR2F4	Fig 1 detail B / 4	$1\frac{3}{8}$ / $\frac{9}{32}$	2 OFF 1 2 1		1 2	4 3							
-022 ISR3F4	Fig 1 detail A / 4	3 / $\frac{9}{32}$	1 2 2 1 OFF	1/2 Blank		1 2 3			4 5 6			7 8 9	1/2 Blank

[1] External jumpers not furnished

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REQUIREMENTS:

Applicable specification.	MIL-S-15291
Dimensions and mounting.	See figure 1
Switching characteristics.	See table I
Angle of throw:	90 degrees
Switching action:	Snap action, reciprocating
Electrical and endurance ratings:	See table II
Stop strength (applicable to switches with stops):	30 inch-pounds
Vibration·	50 Hz, MIL-STD-167-1
Shock (high impact)	High impact, MIL-S-901
Insulation (spacers and decks):	Glass alkyd, MAI-60 (or MAI-30) in accordance with MIL-M-14
Overload·	600 percent of ac rating, 50 make and break operations
Contact resistance:	0 02 ohms maximum
Dielectric withstanding voltage.	1,250 Vrms
Insulation resistance.	200 megohms minimum
Temperature rise:	50 °C maximum
Contact material	Rotor, phosphor bronze – Stationary contact, copper
Mounting screws:	.190-24UNC-2A (4) length and headstyle to suit application. Screws not furnished.

TABLE II. *Electrical and endurance ratings*

Tests	Current (amperes)	Voltage (volts)	Electrical operations (number of operations)	Test rate (operations per minute)	De-energized operations (number of operations)
Alternating current (rms)	10	125	30,000	30	20,000
Direct current	10	120	20,000	15	–

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Terminal marking: Terminal markings shown in table I locate terminals as viewed from the front of the switch (handle end). Markings shall be stamped in front surfaces of terminals, clear of the screw heads.

GENERAL INFORMATION:

Switches not covered by specification sheets: Switches which are fabricated from standard parts, as used in qualified switches, but which do not comply with switches detailed herein with respect to circuit characteristics, switching action, mounting arrangement, and handle details may be acquired under this specification from contractors having qualification approval under this specification.

Extended ratings: Switches detailed herein have been tested and found satisfactory at the extended ratings listed in table III. Reduced life expectancy must be anticipated for switches used at these increased voltage or current levels. Tests under the conditions of table III are not required for qualification acceptance and they are not repeated routinely as for maintenance of qualification. Supplemental evaluations and tests applicable to particular circuit requirements are recommended.

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TABLE III *Extended ratings.*

Switching characteristics	AC - 60 or 400 Hz											
	125 volts				250 volts				500 volts			
	Resistive or lamp load		Inductive load 0.75 p.f.		Resistive or lamp load		Inductive load 0.75 p.f.		Resistive or lamp load		Inductive load 0.75 p.f.	
	Amp	Operations	Amp	Operations	Amp.	Operations	Amp.	Operations	Amp.	Operations	Amp.	Operations
All[1]	10	30,000	10	30,000	---	---	---	---	---	---	---	---
C,E,F[2]	15	10,000	15	10,000	10	10,000	10	10,000	5	10,000	5	10,000
A,B[3]	15	10,000	15	10,000	10	10,000	10	10,000	5	10,000	5	10,000
A,B	15	10,000	15	10,000	15	10,000	15	10,000	10	10,000	10	10,000
	DC											
	120 volts				250 volts				350 volts			
	Resistive or lamp load		Inductive load[4]		Resistive or lamp load		Inductive load[4]		Resistive or lamp load		Inductive load[4]	
	Amp.	Operations	Amp.	Operations	Amp.	Operations	Amp.	Operations	Amp.	Operations	Amp.	Operations
All[1]	5	20,000	---	---	---	---	---	---	---	---	---	---
C,E,F	10	10,000	10	10,000	5	10,000	5	10,000	2	10,000	2	10,000
C,E,F[2]	---	---	---	---	---	---	---	---	---	---	---	---
A,B[2]	10	10,000	10	10,000	5	10,000	5	10,000	2	10,000	2	10,000
A,B	10	20,000	15	10,000	---	---	---	---	---	---	---	---

[1]45 degree blades are rated only at 5 amps, 125 Vac, 120 Vdc

[2]Switching characteristic F is not rated for 500 volts in class 1SR.

[3]Single pole break - all other ratings are based on breaking both sides of the line in accordance with figures 1 and 2 of MIL-S-15291

[4]0.24 henry for inductive circuit

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APPLICATION AND ACQUISITION GUIDE: PIN and type designation cross reference shall be as shown in table IV.

TABLE IV. *Application and acquisition guide*

M15291/3 dash No	Type designation	For new or existing design	For re-placement	Circuit configuration
-001	1SR2A3		X[1]	A (on-off-on-off)
-002	1SR2A4	X		A (on-off-on-off)
-003	1SR2A4A	X		A (off-on)
-004	1SR3A4	X		A (on-off-on-off)
-005	1SR4A3		X[2]	A (on-off-on-off)
-006	1SR4A4	X		A (on-off-on-off)
-007	1SR4A4A	X		A (off-on)
-008	1SR2B4		X[3]	B (on 1-off-on 2-off)
-009	1SR2B4A	X		B (on 1-off-on 2-off)
-010	1SR3B3		X[4]	B (on 1-off-on 2-off)
-011	1SR3B4	X		B (on 1-off-on 2-off)
-012	1SR4B4	X		B (on 1-off-on 2-off)
-013	1SR1C4	X		C (on 1-on 2-on 1-on 2)
-014	1SR2C3		X[5]	C (on 1-on 2-on 1-on 2)
-015	1SR2C4		X[5]	C (on 1-on 2-on 1-on 2)
-016	1SR2C4A	X		C (on 1-on 2-on 1-on 2)
-017	1SR3C4	X		C (on 1-on 2-on 1-on 2)
-018	1SR4C4	X		C (on 1-on 2-on 1-on 2)
-019	1SR2E4	X		E (off-on 1-on 2-on 3)
-020	1SR2F3		X[6]	F (off-on 1-on 1 & 2-on 2)
-021	1SR2F4	X		F (off-on 1-on 1 & 2-on 2)
-022	1SR3F4	X		F (off-on 1-on 1 & 2-on 2)

[1]Use M15291/3-002 in new designs
[2]Use M15291/3-006 in new designs
[3]Use M15291/3-009 in new designs
[4]Use M15291/3-011 in new designs
[5]Use M15291/3-016 in new designs
[6]Use M15291/3-021 in new designs

Revision letters are not used to denote changes due to the extensiveness of the changes

Preparing activity
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